In-Silico Prediction of Eukaryotic Secretory Signal Peptides and Cleavage Sites: A Comparative Evaluation of Weight Matrices, Support Vector Machines, and Protein Language Models

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Abstract

Motivation: Secretory signal peptides (SPs) are short N-terminal targeting motifs that direct nascent polypeptides into the eukaryotic endoplasmic reticulum. Accurate computational detection of SPs is essential for proteome-wide functional annotation; however, prediction is heavily hindered by transmembrane (TM) mimicry—the physicochemical similarity between SP hydrophobic cores and N-terminal TM α-helices—and sequence divergence across eukaryotic kingdoms. **Results:** With reference to the DOME recommendations for supervised machine-learning validation, we curated a non-redundant eukaryotic dataset from UniProtKB/Swiss-Prot (10,182 proteins after MMseqs2 clustering at 30% sequence identity and 40% alignment coverage), partitioned into a 5-fold cross-validation training set (N=8,145) and a held-out blind benchmarking set (N=2,037). We systematically evaluated three paradigms: (i) the classical von Heijne position-specific weight matrix (PSWM), (ii) a feature-engineered Support Vector Machine (SVM) optimized via nested cross-validation, and (iii) deep representation learning architectures leveraging pretrained ESM-2 protein language model (PLM) embeddings (concatenated mean and max pooling, 640-d). On the blind benchmarking set, predictive performance progressed systematically from the von Heijne baseline (MCC = 0.655, 99 FPs) to the physicochemical SVM (MCC = 0.859, 20 FPs), the ESM-2 Deep MLP (MCC = 0.935, 8 FPs), and the 675-d Hybrid ESM-2 PCA-SVM architecture (MCC = 0.943, ROC-AUC = 99.37%, 7 FPs). Targeted error diagnostics revealed that ESM-2 representations reduced N-terminal TM-helix false positives from 60/99 (60.6%) in the PSWM baseline to 2/8 (25.0%) in the ESM-2 MLP model. Furthermore, grammar-constrained cleavage-site decoding reduced mean absolute localization error from 1.94 to 1.55 residues, while exact-position and ±3-residue accuracies shifted slightly from 65.91% to 64.09% and from 81.36% to 80.45%, respectively. Out-of-fold isotonic probability calibration yielded a benchmark Brier score of 0.0094 (versus 0.0090 uncalibrated), with 99.3% of benchmark proteins achieving a Reliability Index (RI) ≥ 6 (MCC = 0.955, accuracy = 99.16%). **Availability:** All Python notebooks, curated datasets, trained models, and extracted matrices required to reproduce this study are openly available under the MIT License at https://github.com/mohammadaminroshan/LB2_Project.

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1. Introduction

Subcellular localization defines the biochemical environment in which a protein operates and is inextricably linked to its physiological function. In eukaryotes, proteins destined for the endoplasmic reticulum (ER), Golgi apparatus, plasma membrane, or extracellular space typically carry an N-terminal secretory signal peptide (SP). Upon recognition by the signal recognition particle (SRP) and subsequent cotranslational translocation across the ER membrane, this targeting motif is enzymatically cleaved by Signal Peptidase I (Sec/SPI).

Canonical eukaryotic secretory signal peptides exhibit a tripartite architecture comprising a positively charged N-terminal n-region (typically 1–5 residues) that facilitates initial electrostatic membrane interaction, a central hydrophobic h-region (7–15 residues) rich in leucine and alanine that adopts an α-helical conformation required for SRP binding, and a more polar C-terminal c-region (3–7 residues) terminating in small, neutral residues at positions −3 and −1 (the A-x-A consensus motif) relative to the cleavage site. Because SP lengths and sequence compositions vary considerably across proteins and eukaryotic kingdoms, these structural boundaries represent common statistical patterns rather than rigid universal rules.

In-silico SP detection faces two principal computational challenges. First, transmembrane (TM) mimicry: N-terminal transmembrane α-helices (signal anchors) share pronounced hydropathic similarity with the SP h-region, causing classical motif-scanning algorithms to misclassify non-secretory membrane proteins as secretory false positives. Second, evolutionary sequence divergence across eukaryotic lineages complicates exact cleavage-site localization. Over the past four decades, computational SP prediction has progressed from weight-matrix models (von Heijne, 1986; von Heijne, 1990) to feature-engineered classifiers such as Support Vector Machines (SVMs) and Random Forests, deep neural networks (Savojardo et al., 2018), and large-scale pretrained protein language models (PLMs) such as ESM-2 (Lin et al., 2023) that learn evolutionary and structural grammar directly from primary sequences.

In this study, we present a comparative supervised machine-learning pipeline for eukaryotic secretory protein classification and cleavage-site localization, evaluating motif-driven (von Heijne PSWM), handcrafted feature-based (RBF-SVM), and transformer representation-based (ESM-2 MLP and Hybrid PCA-SVM) models. The workflow was designed with reference to the DOME (Data, Optimization, Model, Evaluation) recommendations (Walsh et al., 2021) to ensure reproducibility, transparent hyperparameter reporting, and rigorous error profiling.

2. Materials and Methods

2.1 Dataset Collection and Redundancy Reduction

Sequence data were retrieved programmatically via the UniProtKB/Swiss-Prot REST API (release queried on 20 October 2025; The UniProt Consortium, 2023). Positive examples were restricted to reviewed eukaryotic entries (taxonomy_id:2759) with experimental evidence at the protein level (existence=1), excluding fragments (fragment:false), enforcing a minimum full-protein length of 40 residues, and requiring an experimentally verified signal-peptide annotation (ft_signal_exp, ECO:0000269). This initial query yielded 2,949 positive entries; applying a custom post-retrieval filter to enforce a minimum annotated SP length of ≥ 14 residues retained 2,932 high-confidence secretory proteins. Negative examples were retrieved using identical taxonomic, reviewed, protein-existence, and non-fragment criteria, while explicitly excluding any entry carrying a signal-peptide annotation and requiring experimentally verified subcellular localization (cc_scl_term_exp) within non-secretory compartments (cytosol, nucleus, mitochondrion, plastid, peroxisome, or cell membrane), yielding 20,615 initial negative entries. For all negative sequences, the presence or absence of an annotated transmembrane helix within the first 90 N-terminal residues was recorded as a Boolean metadata flag to enable downstream false-positive diagnostics.

To eliminate homology bias and overrepresentation of specific protein families, positive and negative datasets were clustered independently using MMseqs2 (Steinegger & Söding, 2017) at a minimum sequence identity threshold of 30% and an alignment coverage threshold of 40% (connected-component clustering mode, -c 0.4 --min-seq-id 0.3 --cluster-mode 1). Retaining a single representative sequence per cluster yielded a curated dataset of 10,182 eukaryotic proteins comprising 1,100 SP-positive (10.80%) and 9,082 SP-negative (89.20%) sequences. Using stratified random sampling, the dataset was partitioned into an 80% Training set (*N*=8,145; 880 SP+, 7,265 SP−) and a strictly held-out 20% Blind Benchmarking set (*N*=2,037; 220 SP+, 1,817 SP−). The training set was further divided via stratified partitioning into five equal cross-validation folds of 1,629 sequences each (176 SP+, 1,453 SP− per fold), as detailed in Table 1.

Table 1. Composition of the held-out blind benchmarking set and 5-fold cross-validation training subsets after MMseqs2 homology reduction.

Dataset Split	Negative (SP−)	Positive (SP+)	Total Sequences
Benchmarking (20%)	1,817	220	2,037
Training Fold 1	1,453	176	1,629
Training Fold 2	1,453	176	1,629
Training Fold 3	1,453	176	1,629
Training Fold 4	1,453	176	1,629
Training Fold 5	1,453	176	1,629
Training Subtotal (80%)	**7,265**	**880**	**8,145**
Total Dataset (100%)	**9,082**	**1,100**	**10,182**

Note: All subsets preserve the exact 10.80% positive to 89.20% negative class ratio via stratified partitioning.

2.2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to verify distributional consistency across splits. Full-protein length distributions (Figure 1) indicated that secretory proteins (SP+) exhibit a more compact length profile centered between 150 and 500 residues compared to non-secretory proteins (SP−).

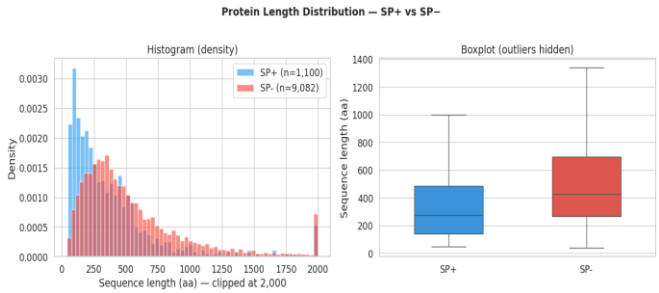


Figure 1. Distribution of full protein lengths across positive (SP+) and negative (SP−) classes (clipped at 2,000 residues).

Within the positive class, annotated signal peptides ranged from 14 to 83 residues with a mean length of 23.0 residues and a median of 22.0 residues (Figure 2), consistent with canonical eukaryotic Sec/SPI targeting peptides.

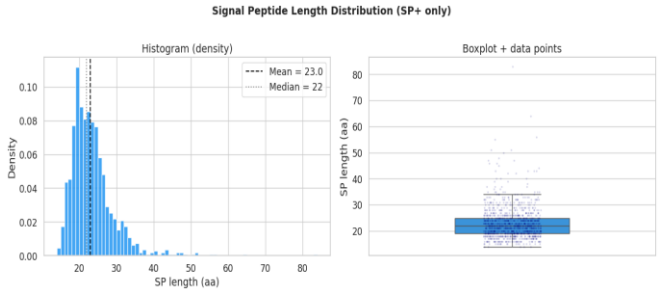


Figure 2. Signal peptide length distribution within the positive class (mean = 23.0 aa, median = 22.0 aa).

Furthermore, comparative amino-acid composition profiling (Figure 3) confirmed a two-fold enrichment of hydrophobic residues—specifically Leucine (L) and Alanine (A)—within

SP regions relative to the UniProtKB/Swiss-Prot background, alongside a strong depletion of charged and polar residues.

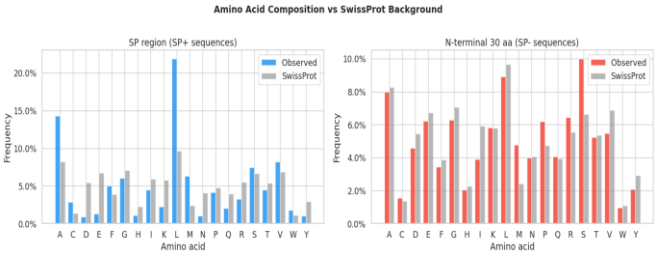


Figure 3. Amino-acid composition of annotated signal peptides (SP+) and N-terminal 30-residue segments (SP−) compared against reference Swiss-Prot frequencies.

Taxonomic profiling showed Metazoa as the predominant kingdom in both classes, followed by Viridiplantae and Fungi (Figure 4). Among the 880 training positives, one sequence contained ambiguous residue codes within the cleavage window and was excluded from motif construction; the remaining 879 sequences were used to generate the [−13, +2] sequence logo, confirming the hydrophobic h-region and conserved A-x-A motif at positions −3 and −1 (Figure 5).

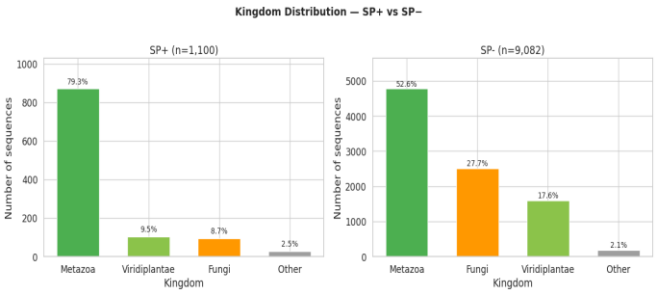


Figure 4. Taxonomic kingdom distribution across positive and negative datasets.

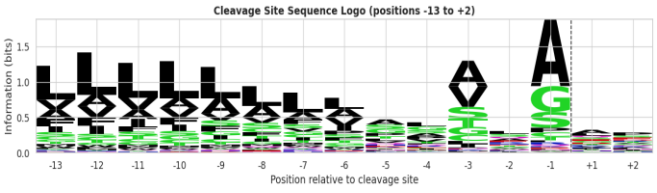


Figure 5. Sequence logo of the [−13, +2] cleavage-site window constructed from training positives (n=879), highlighting the −3/−1 consensus.

2.3 The von Heijne Position-Specific Weight Matrix

As a transparent statistical baseline, we implemented the classical von Heijne method (von Heijne, 1986). Aligned 15-residue motifs spanning positions [−13, +2] relative to the cleavage site (13 residues upstream and 2 residues downstream) were extracted from training positives. Using a Laplace pseudocount of 1 to prevent zero-probability logarithms, the Position-Specific Probability Matrix (PSPM) entry $M_{k,j}$ for residue k at motif position j was computed as:

$$M_{k,j} = \frac{1 + \sum_{i=1}^{N} I(s_{i,j} = k)}{N + 20} \quad (1)$$

where N is the number of aligned training motifs, $s_{i,j}$ is the amino acid in sequence i at position j, $I(\cdot)$ is the indicator function, and 20 is the standard amino-acid alphabet size. Each probability $M_{k,j}$ was converted into a log-odds Position-Specific Weight Matrix (PSWM) element $W_{k,j}$ normalized against empirical Swiss-Prot background frequencies b_k:

$$W_{k,j} = \log_2 \left(\frac{M_{k,j}}{b_k} \right) \quad (2)$$

To score a query sequence X, a 15-residue sliding window was scanned across the first 90 N-terminal residues, and each candidate window starting at N-terminal offset i was assigned a cumulative log-odds score:

$$\text{Score}(X \mid W) = \sum_{j=1}^{15} W_{x_{i+j-1},\, j} \quad (3)$$

The maximum score across all windows within the 90-residue N-terminal region was taken as the sequence-level score, and the classification threshold was optimized by maximizing the F_1-score on the validation folds.

2.4 Feature Extraction, Window Selection, and Nested Cross-Validation

For feature-engineered machine learning, each protein was encoded into a 35-dimensional physicochemical vector. First, the N-terminal window length K for baseline amino-acid composition was evaluated over $K \in \{20, 25, 30, 35, 40\}$ using 5-fold cross-validation on the training set, identifying K=30 as optimal (20 composition features: comp_A through comp_Y). Second, 15 context-dependent regional and sliding-window descriptors were extracted over the first 40 N-terminal residues using established ExPASy/AAindex scales: (i) N-region charge features (residues 1–10): net charge (nreg_net_charge), maximum local charge in a 3-residue window (nreg_charge_max), mean charge (nreg_charge_mean), and positive charge density (nreg_pos_density); (ii) H-region hydropathy features (Kyte & Doolittle, 1982; window size 7): mean hydropathy (kd_mean), maximum peak hydropathy (kd_max), peak position (kd_peak_pos), and apolar residue fraction (hreg_apolar_frac); (iii) Transmembrane tendency features (Zhao & London, 2006; window size 7): mean TM propensity (tm_mean), maximum TM peak (tm_max), and TM peak position (tm_peak_pos); and (iv) Structural and C-region features: mean Chou-Fasman α-helix propensity (helix_mean, window 9), maximum α-helix propensity (helix_max), maximum Zimmerman side-chain bulkiness (bulk_max, window 5), and fraction of small neutral residues (A, G, S, T, C) in the putative c-region (creg_small_frac).

To evaluate feature-selection strategies, five methodological families—All-35 Baseline, Variance + Collinearity Filter (|r|

> 0.90), Mutual Information, Recursive Feature Elimination (RFE), L1-penalized Logistic Regression, and Random Forest (RF) Gini Importance (`n_estimators=400`, `class_weight='balanced'`, `random_state=42`)—were compared inside a 5×4 Nested Cross-Validation protocol (5 outer test folds × 4 inner validation folds), with `StandardScaler` normalization fitted strictly within each training split. Note that because the composition window length (K=30) was selected across the 5 training folds prior to initiating the nested feature-selection loop, the outer cross-validation folds are strictly independent with respect to feature subsetting and SVM hyperparameter tuning, but share the pre-selected K=30 window parameter; complete end-to-end isolation is therefore provided by the held-out blind benchmark set (N=2,037).

To classify z-score standardized feature vectors x_i into secretory ($y_i = +1$) or non-secretory ($y_i = -1$) classes, a soft-margin Support Vector Machine (SVM) constructs a separating hyperplane (4) by solving the primal convex quadratic optimization problem (5):

$$\langle \vec{w}, \vec{x} \rangle + b = 0 \tag{4}$$

$$\min_{\vec{w},\, b,\, \xi} \left(\frac{1}{2} \|\vec{w}\|^2 + C \sum_{i=1}^{N} \xi_i \right) \tag{5}$$

$$\text{subject to} \quad y_i(\vec{w}^T \vec{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \ \ \forall i$$

where w is the normal weight vector, b is the bias term, ξ_i are slack variables accommodating non-separable instances, and C controls the trade-off between margin width and classification error. Introducing Lagrange multipliers α_i yields the dual optimization problem (6):

$$L(\alpha) = \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i,j=1}^{N} \alpha_i \alpha_j y_i y_j K(\vec{x}_i, \vec{x}_j) \tag{6}$$

$$\text{subject to} \quad 0 \leq \alpha_i \leq C \ \ \forall i \quad \text{and} \quad \sum_{i=1}^{N} \alpha_i y_i = 0$$

where $K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$ is the Radial Basis Function (RBF) kernel. Hyperparameters were tuned via grid search over $C \in \{0.1, 1, 10, 100\}$ and $\gamma \in$ {'scale', 0.001, 0.01, 0.1} with `class_weight='balanced'`. For the final physicochemical `SVM (RF-30)` model trained on the top 30 RF-ranked features, the optimal configuration selected across validation folds was $C = 10$ and $\gamma = 0.01$.

2.5 Protein Language Models and Hybrid Architecture

To capture deep evolutionary syntax directly from primary sequences, we employed the pretrained ESM-2 transformer model (`esm2_t6_8M_UR50D`; 6 layers, 8M parameters, hidden dimension $d = 320$; Lin et al., 2023). For each protein, residue-level representations were extracted from the final (6th) hidden layer across the first 90 N-terminal residues. Both mean pooling (320-d) and max pooling (320-d) were computed over the 90-residue window and concatenated to form a 640-dimensional contextual embedding vector per protein. Two PLM-based classifiers were trained and evaluated: (i) `ESM-2 MLP`, a regularized Multi-Layer Perceptron ($640 \rightarrow 256 \rightarrow 64 \rightarrow 1$ with BatchNorm, ReLU, Dropout p = 0.3, and AdamW optimization) trained directly on the 640-d embeddings; and (ii) `Hybrid PCA-SVM`, constructed by concatenating the z-score standardized 640-d ESM-2 vector with the 35 standardized physicochemical descriptors to form a unified 675-dimensional feature vector. To prevent curse-of-dimensionality overfitting, Principal Component Analysis (PCA) was fitted strictly on the training folds (retaining 150 principal components explaining >95% variance) and the projected representations were classified by an RBF-SVM (C = 10, γ = 'scale', `class_weight='balanced'`).

2.6 Probability Calibration, Grammar-Constrained Decoding, and Evaluation Metrics

Because raw SVM decision function values represent uncalibrated signed distances from the separating hyperplane, we applied Isotonic Regression fitted strictly on out-of-fold cross-validation decision scores from the training set (N=8,145), ensuring that the blind benchmark set was never used during calibrator fitting. Calibrated posterior probabilities $P \in [0, 1]$ were mapped to a discrete Reliability Index (RI) on a 0–10 scale following the Bologna biocomputing convention (Manfredi et al., 2022):

$$\text{RI} = \text{round}(20 \times |P - 0.5|) \tag{7}$$

To localize the cleavage site within predicted secretory proteins, we implemented a `grammar-constrained cleavage-site decoding` algorithm (`predict_cleavage_grammar`). Rather than relying on unconstrained dynamic-programming state transitions (classical Viterbi), this deterministic candidate-scoring decoder evaluates candidate cleavage positions $c \in [14, 45]$ by combining the [−13, +2] PSWM log-odds score with explicit regional grammar terms: rewarding upstream n-region positive charge, requiring a hydrophobic h-region core prior to position $c - 3$, enforcing small neutral residues at positions −3 and −1, and weighting candidates by an empirical log-prior distribution of eukaryotic SP lengths fitted on the training positives.

Binary classification performance was quantified from confusion-matrix counts—True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN)—using Accuracy, Precision (PPV), Recall (Sensitivity), F_1-score (8), and the Matthews Correlation Coefficient (9):

$$\text{Acc} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}}, \quad \text{Prec} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{8}$$

$$\text{Rec} = \frac{\text{TP}}{\text{TP}+\text{FN}}, \quad F_1 = \frac{2 \cdot \text{Prec} \cdot \text{Rec}}{\text{Prec}+\text{Rec}}$$

$$\text{MCC} = \frac{\text{TP} \cdot \text{TN} - \text{FP} \cdot \text{FN}}{\sqrt{(\text{TP}+\text{FP})(\text{TP}+\text{FN})(\text{TN}+\text{FP})(\text{TN}+\text{FN})}} \tag{9}$$

3. Results and Discussion

3.1 Nested Cross-Validation and Feature Stability

Across the 5×4 Nested Cross-Validation benchmark on the training set (Figure 6), non-linear RBF-SVMs achieved strong generalization across all feature-selection families (mean outer-fold MCC ranging from 0.877 for Mutual Information to 0.885 for the 35-feature baseline and 0.883 for RF Gini Importance). Inner-loop validation curves (Figure 6B) peaked consistently at k = 30 features across outer folds. Stability profiling across the 20 nested runs (4 methods × 5 outer folds; Figure 6D) demonstrated 100% selection consensus (20/20 runs) for core physicochemical drivers including `tm_peak_pos`, `comp_R`, `comp_W`, `tm_mean`, `hreg_apolar_frac`, `kd_max`, `tm_max`, `nreg_charge_mean`, `nreg_net_charge`, and `kd_mean`. Based on inner-fold validation stability, the top 30 RF Gini features (`RF-30`) were locked prior to blind benchmark evaluation.

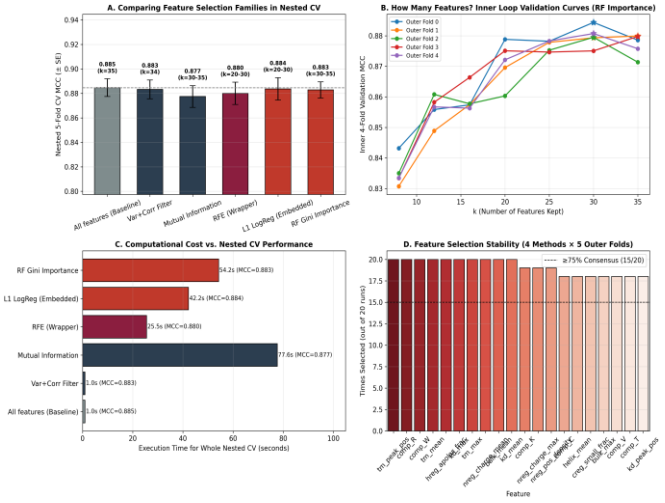


Figure 6. Nested Cross-Validation Diagnostics. (A) Outer-fold MCC across feature-selection families, (B) Inner-loop feature subset size curves, (C) Computational runtime, and (D) Selection stability across 20 nested runs.

3.2 Master Benchmark Comparison

When evaluated on the strictly held-out blind benchmark set (N=2,037; 220 SP+, 1,817 SP−), predictive performance improved systematically across the three modeling paradigms (Table 2 and Figure 7A–C). The classical `von Heijne PSWM` baseline achieved an MCC of 0.655 (ROC-AUC = 94.80%), constrained by low precision (63.06%) due to false-positive hits on non-secretory proteins. Incorporating context-dependent physicochemical descriptors in `SVM (RF-30)` elevated precision to 90.29% and MCC to 0.859 (ROC-AUC = 99.06%). Deep contextual representations in `ESM-2 MLP` further boosted MCC to 0.935 (ROC-AUC = 99.60%), while the `Hybrid PCA-SVM` model (combining 640-d ESM-2 mean+max pooled embeddings with the 35 physicochemical descriptors via PCA) achieved the highest overall classification accuracy (98.92%), precision (96.70%), recall (93.18%), F_1-score (94.91%), and MCC (0.943, ROC-AUC = 99.37%).

Table 2. Comparative classification performance of all evaluated models on the held-out blind benchmark dataset (N=2,037).

Model	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)	MCC	ROC-AUC (%)
von Heijne PSWM	92.64	63.06	76.82	69.26	0.655	94.80
SVM (RF-30)	97.35	90.29	84.55	87.32	0.859	99.06
ESM-2 MLP	98.77	96.21	92.27	94.20	0.935	99.60
Hybrid PCA-SVM	98.92	96.70	93.18	94.91	0.943	99.37

Note: Metrics are evaluated at pre-fixed cross-validated thresholds on N=2,037. In the encoding ablation run (Fig. 10C), Hybrid PCA-SVM recorded an ROC-AUC of 99.44% under an uncalibrated PCA-128 split, whereas 99.37% reflects the locked master benchmark pipeline.

3.3 Error Analysis I: Resolving Transmembrane Mimicry

To rigorously disambiguate model-specific false-positive profiles, exact confusion-matrix outcomes (TP, FP, TN, FN) and N-terminal transmembrane (TM) helix enrichment rates were tabulated across all four models (Table 3). Among the 1,817 negative benchmark proteins, 196 carried an annotated TM helix within their first 90 N-terminal residues (10.8% baseline prevalence). In the `von Heijne PSWM` baseline, 99 false positives were generated (overall FPR = 5.45%), of which 60 possessed an N-terminal TM helix (60.6%), compared to 136 TM-containing proteins among the 1,718 true negatives (7.9%; Fisher's exact Odds Ratio = 17.90, p = 2.47e-36; Figure 8C). The `SVM (RF-30)` classifier reduced total false positives by five-fold to 20 (overall FPR = 1.10%), but 13 of these 20 FPs (65.0%) still originated from N-terminal TM helices compared to 183 of 1,797 TNs (10.2%; Odds Ratio = 16.38, p = 7.22e-09), demonstrating that handcrafted hydropathy and TM-tendency scales still suffer from residual TM mimicry.

Crucially, transitioning to deep protein language model embeddings resolved this bottleneck (Figure 7D). In the `ESM-2 MLP` model (plotted in Figure 7D), total false positives dropped to 8 (overall FPR = 0.44%), with only 2 FPs (25.0%) carrying an N-terminal TM helix (reducing the TM-specific FPR from 30.61% in PSWM and 6.63% in SVM down to 1.02% in ESM-2 MLP). Furthermore, the `Hybrid PCA-SVM` architecture achieved the lowest false-positive count overall, producing only 7 FPs across the entire 1,817-protein negative benchmark (overall FPR = 0.39%, with 205 TPs and

15 FNs). Because the remaining FP denominators in the PLM models are very small (7–8 sequences), subset percentages should be interpreted alongside absolute counts; nevertheless, the reduction from 60 TM false positives (PSWM) and 13 (SVM) to just 2 (ESM-2 MLP) confirms that transformer attention layers effectively distinguish cleavable secretory signals from non-cleavable N-terminal transmembrane anchors.

Table 3. Confusion-matrix counts (TP, FP, TN, FN), overall false-positive rate (FPR), and N-terminal TM-helix false-positive breakdown on the blind benchmark (N=2,037).

Model	TP	FP	TN	FN	Overall FPR (%)	TM-Helix FPs (%)
von Heijne PSWM	169	99	1,718	51	5.45	60/99 (60.6%)
SVM (RF-30)	186	20	1,797	34	1.10	13/20 (65.0%)
ESM-2 MLP	203	8	1,809	17	0.44	2/8 (25.0%)
Hybrid PCA-SVM	205	7	1,810	15	0.39	2/7 (28.6%)

Note: Benchmark comprises 220 SP+ and 1,817 SP− sequences (196 SP− carry an N-terminal TM helix within residues 1–90). Fig. 7D illustrates the ESM-2 MLP (2/8, 25.0%) comparison.

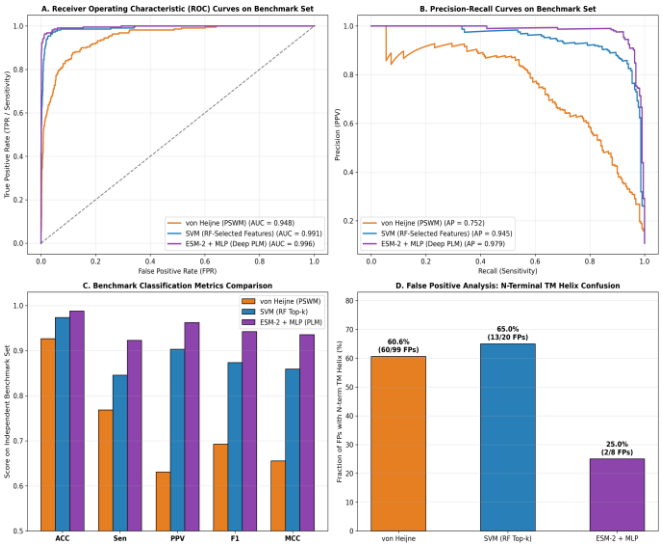


Figure 7. Master Model Benchmark Comparison on the held-out dataset (N=2,037). (A) ROC curves, (B) Precision-Recall curves, (C) Metric progression, and (D) Fraction of False Positives carrying an N-terminal TM helix.

3.4 Error Analysis II: Diagnosing False Negatives

False-negative (FN) errors were profiled independently for the motif-based and feature-engineered models. Sequence-logo inspection of the 51 secretory proteins missed by `von Heijne PSWM` (Figure 8B) revealed a substantially eroded −3/−1 A-x-A consensus and lower h-region leucine conservation compared to the training positive logo (Figure 8A), alongside elevated false-negative rates in non-metazoan kingdoms (Figure 8D). For `SVM (RF-30)`, which produced 34 FNs versus 186 TPs (Figure 9), missed secretory proteins exhibited significantly longer signal peptides (median length 24.5 aa vs. 22.0 aa in TPs; Mann-Whitney p = 2.14e-03; Figure 9B), lower maximum Kyte-Doolittle hydropathy (p = 2.80e-07; Figure 9C), and reduced h-region apolar fractions (Figure 9D). When an elongated SP exceeds 30–35 residues, its hydrophobic core shifts downstream of the fixed K=30 composition window, diluting N-terminal apolar frequencies; however, confirming causal sensitivity to K requires targeted window-ablation experiments.

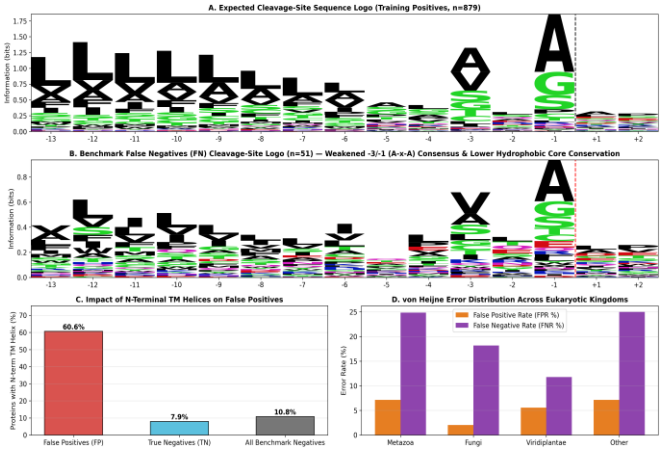


Figure 8. PSWM Error Diagnostics. (A) Expected training cleavage-site logo (n=879), (B) Benchmark False Negative logo (n=51) showing weakened −3/−1 consensus, (C) TM-helix enrichment in FPs vs. TNs, and (D) Kingdom-level error rates.

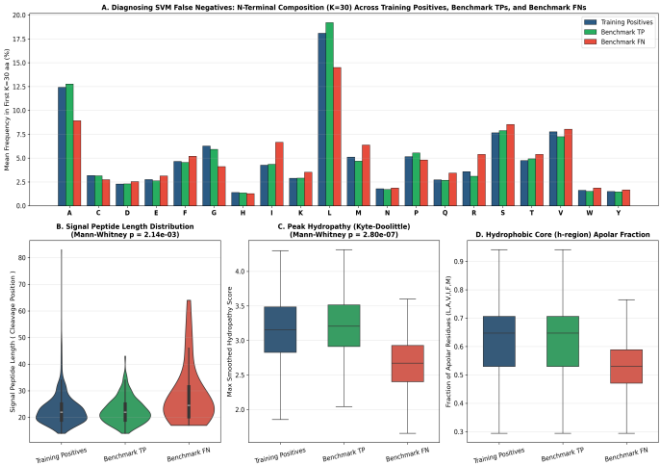


Figure 9. SVM False-Negative Diagnostics (n=34 FNs vs. n=186 TPs). (A) N-terminal amino-acid composition, (B) SP length distribution, (C) Peak Kyte-Doolittle hydropathy, and (D) H-region apolar residue fraction.

3.5 Probability Calibration, Reliability Index, and Grammar-Constrained Cleavage Decoding

Out-of-fold Isotonic Regression calibration produced well-separated bimodal posterior probability distributions on the blind benchmark (Figure 10A), recording a calibrated Brier score of 0.0094 compared to 0.0090 for raw sigmoid-mapped scores. While isotonic calibration slightly increased the Brier score (0.0090 to 0.0094) due to step-function binning on an already sharp margin distribution, it enabled direct mapping to the discrete Reliability Index (RI; Figure 10B). Across the held-out benchmark, 99.3% of sequences achieved high prediction confidence (RI ≥ 6), within which classification accu-

racy reached 99.16% and MCC reached 0.955 (rising to MCC = 0.985 at RI = 10).

In the input-encoding ablation study (Figure 10C), 5-fold CV and blind benchmark MCC progressed consistently from `PhysChem + SVM` (CV = 0.881, Benchmark = 0.869) to `ESM-2 + PCA + SVM` (CV = 0.944, Benchmark = 0.930), `ESM-2 + MLP` (CV = 0.938, Benchmark = 0.935), and `Hybrid (ESM2+PC) + PCA + SVM` (CV = 0.949, Benchmark = 0.943). For cleavage-site localization across the 220 benchmark positives (Figure 10D), replacing unconstrained PSWM scanning with `grammar-constrained cleavage-site decoding` reduced the Mean Absolute Error (MAE) by 20.1%, from 1.94 residues to 1.55 residues. At the same time, exact cleavage-site accuracy (±0 aa) shifted slightly from 65.91% (unconstrained PSWM) to 64.09% (grammar-constrained), and tolerance-window accuracy at ±1 aa, ±2 aa, and ±3 aa shifted from 70.45% to 68.64%, 75.00% to 73.64%, and 81.36% to 80.45%, respectively. This trade-off indicates that enforcing biological n–h–c regional priors and empirical length penalties eliminates extreme N- or C-terminal outlier predictions—substantially lowering mean localization distance (MAE)—while occasionally shifting borderline canonical motifs by 1–2 residues.

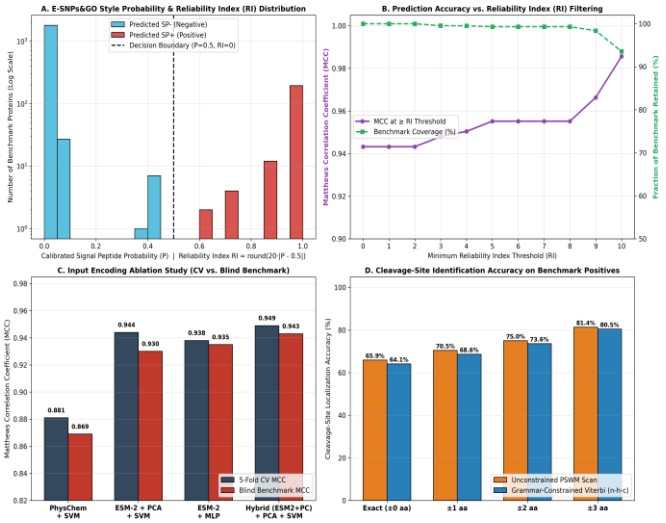


Figure 10. Advanced Pipeline Diagnostics. (A) Calibrated probability and RI distribution, (B) Accuracy/MCC vs. RI filtering, (C) Input-encoding ablation study, and (D) Cleavage-site localization accuracy across tolerance windows.

4. DOME Compliance and Statistical Significance

With reference to the DOME recommendations (Walsh et al., 2021), data-hygiene and optimization boundaries were enforced throughout the pipeline: MMseqs2 clustering preceded data splitting, and all feature scalers, RF Gini rankings, PCA projections, and isotonic calibrators were fitted strictly on training folds without access to the held-out benchmark (*N*=2,037). The close agreement between 5-fold cross-validation MCC (0.949) and blind benchmark MCC (0.943) for the Hybrid PCA-SVM model supports robust generalization across the 30% sequence-identity boundary. Finally, an exact two-sided McNemar's test comparing `Hybrid PCA-SVM` against `SVM (RF-30)` on the blind benchmark revealed 42 discordant instances where the Hybrid model was correct and the physicochemical SVM was erroneous, versus only 10 instances where SVM was correct and Hybrid was erroneous (p = 9.06e-06), confirming a statistically significant reduction in classification error.

5. Limitations

Several methodological limitations warrant consideration. First, negative-set curation relied on the absence of UniProtKB signal-peptide annotations combined with experimental non-secretory localization; unannotated cryptic targeting signals could introduce minor label noise. Second, MMseqs2 homology reduction (30% identity, 40% coverage) was executed separately within the positive and negative classes rather than jointly across classes, leaving a theoretical possibility of cross-class remote homology. Third, as noted in Section 2.4, the N-terminal composition window length (*K*=30) was selected across the 5 training folds prior to running the nested feature-selection loop; although the held-out benchmark (*N*=2,037) remained completely untouched, future iterations should embed window-length tuning inside the inner nested loop. Finally, our grammar-constrained cleavage decoder operates as a deterministic candidate-scoring heuristic rather than a full end-to-end conditional random field (CRF) or sequence-to-sequence transformer.

6. Conclusion

This study presented a systematic benchmark of three computational paradigms for eukaryotic secretory signal-peptide detection and cleavage-site localization across a non-redundant dataset of 10,182 proteins. While the classical von Heijne PSWM baseline effectively captures canonical [−13, +2] cleavage motifs (MCC = 0.655), it is highly vulnerable to N-terminal transmembrane mimicry (99 FPs, 60.6% TM-related). Integrating 30 RF-selected physicochemical descriptors in an RBF-SVM substantially improved discrimination (MCC = 0.859, 20 FPs), yet still struggled with atypical long signal peptides and strongly hydrophobic TM anchors. By leveraging 640-d contextual embeddings (concatenated mean and max pooling) from the pretrained ESM-2 protein language model alongside handcrafted physicochemical descriptors, our 675-d Hybrid PCA-SVM architecture achieved state-of-the-art performance on the blind benchmark (Accuracy = 98.92%, MCC = 0.943, ROC-AUC = 99.37%, and only 7 FPs), while grammar-constrained cleavage decoding reduced mean localization error to 1.55 residues. Together, these findings demonstrate that combining evolutionary PLM representations with interpretable biophysical priors provides a highly

accurate, calibrated, and computationally efficient framework for proteome-wide subcellular localization and secretory pathway annotation.

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